APC 523 PSET 2

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1 Polynomial Interpolation

Implement a method using Lagrange polynomial interpolation of degree p using for the p+1 node points the roots of the degree p + 1 Chebyshev polynomial. Write a function that takes as input f(x) on the interval [-1,1] and interpolates it to g(x). Compute the L2 error and the maximum error of your interpolation using a fine grid with N = 1000 points uniformly distributed in the domain.

Using the interval $x \in [-1, 1]$ *and p + 1 Chebyshev node points, interpolate these three functions:*

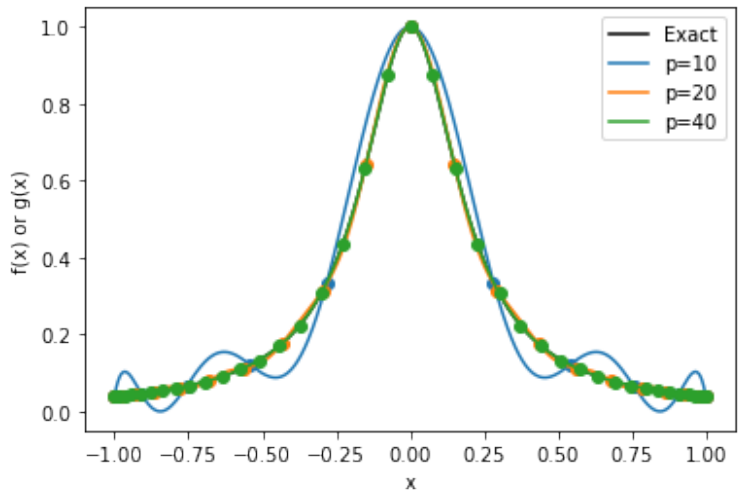
(a) $f(x) = 1/(1 + 25x^2)$

(b) $f(x) = |x|$

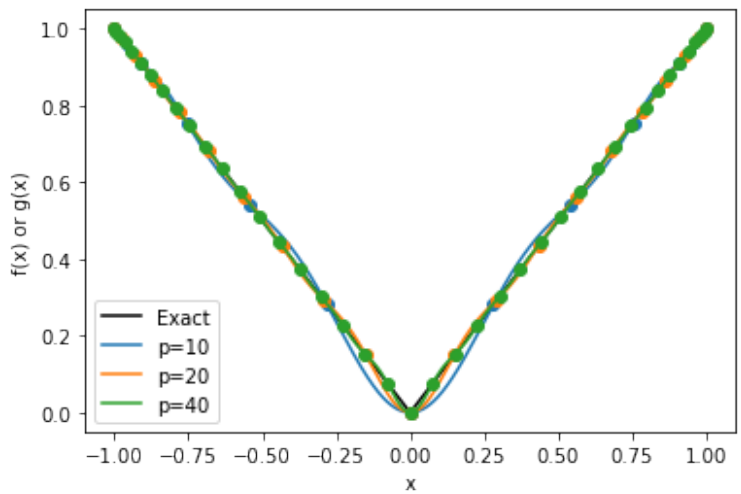
(c) $f(x) = H(x - 0.25)$

where $H(x)$ *is the step function on x. Using only 2 figures,*

(1) *Plot the actual function f(x) and its approximation g(x) as function of x on the fine grid. Do this for each function above using 3 different Lagrange polynomial degrees p = {10, 20, 40}. Mark clearly the p + 1 Chebyshev nodes on the plot of g(x) using symbols*


Exact
p=10
p=20
p=40
f(x) or g(x)
x

(a)


Exact
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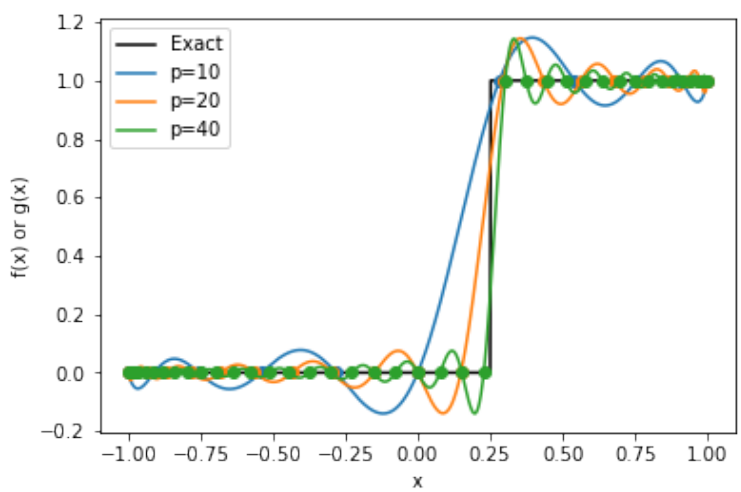
p=20
p=40
f(x) or g(x)
x

(b)


Exact
p=10
p=20
p=40
f(x) or g(x)
x

(c)

(2) *Plot the L2 and maximum errors as a function of $p \in [1, 256]$ using a log-log scale. Label the observed power law dependencies using dashed lines. Interpret the results.*

The L2 and max norm errors for function (a) suggest an exponential dependency on p of the form $A \cdot B^{-p}$, where $B \approx 1.2$ to 1.3 (Figure 2). This is consistent with the expected qualitative results, where Chebyshev interpolants of smooth analytic functions converge geometrically (i.e., proportional to B^{-p} for a constant B) [3].

Functions (b) and (c), on the other hand, are not smooth. Though, away from the discontinuities, both functions behave as polynomials. Thus, Lagrange polynomials can well-approximate the functions away from the discontinuities, meaning that the observed errors emerge almost entirely from the discontinuities.

Lagrange interpolation at Chebyshev nodes of function (b) yields approximately linear convergence (i.e., proportional to p^B where $B \approx -1.0$), as the error directly relates to resolution at the slope (order-1) discontinuity.

Function (c) has an order-0 discontinuity, leading to slower interpolant convergence with p. Indeed, the power law fits from L2 and max norms suggest proportionality to p^B for $B \approx -0.01$ to -0.4 (Figure 3).

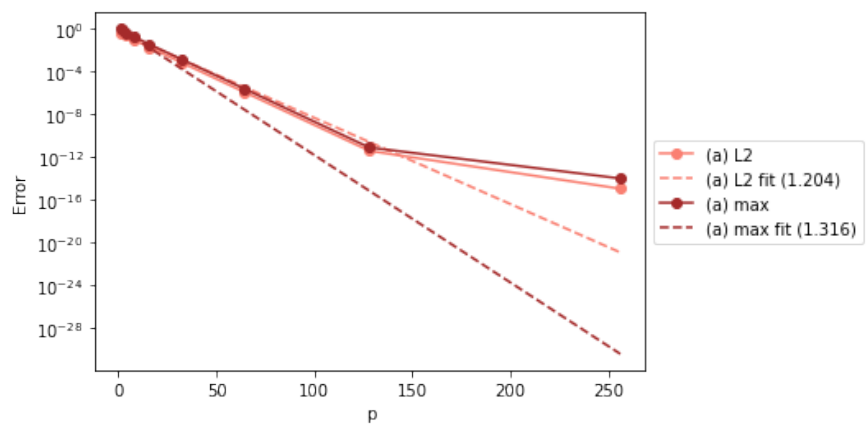


Figure 2: Semi-log y plot for function (a) $f(x) = 1/(1 + 25x^2)$, with dashed lines indicating an exponential fit $g(x) = A \cdot B^{-x}$ and the value B given in parentheses in the legend.

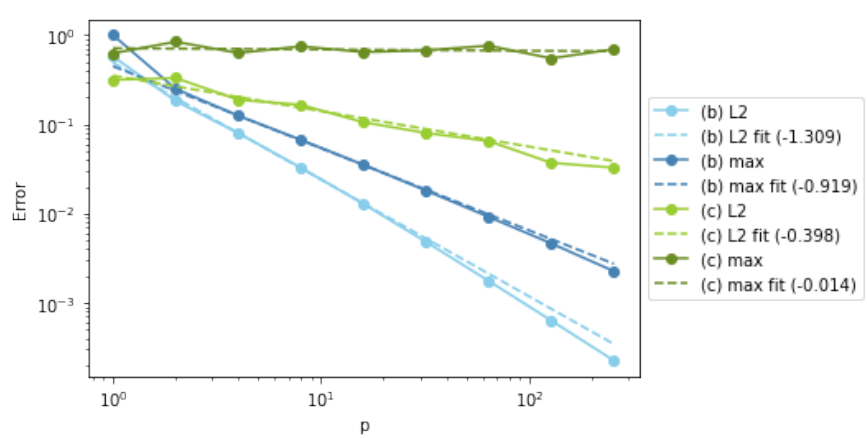


Figure 3: Log-log plot for functions (b) and (c), with dashed lines indicating power law fits of the form $g(x) = A \cdot x^B$, with values of B given in parentheses in the legend.

2 Integration

Write a numerical integration function that takes the following arguments: a function f(x), a number of discrete intervals N, and the integration bounds a and b. The function must return the value of $\int_a^b f(x)dx$. *The integrator should use 5-point Gaussian Quadrature method in each of the N intervals. Plot the relative error of integration as a function of h = (b - a)/N for*

(a) $a = -1$, $b = 1$, $f(x) = x^8$

(b) $a = -1$, $b = 1$, $f(x) = |x - 1/\sqrt{2}|^3$

(c) $a = -1$, $b = 1$, $f(x) = H(x - 1/\sqrt{2})$

(d) $a = 0$, $b = 1$, $f(x) = 1/\sqrt{x}$

Determine the power law dependency of each error curve as a function of h (e.g. $h^{0.5}$, h^1, *or* h^2*). Then, plot the corresponding power-law trend as a dashed line for reference. Explain the power law dependencies that you find for each function in the write-up.*

Five-point Gaussian quadrature integrates polynomials of degree ≤ 9 to floating point precision ($\leq 2p+1$ where $p = 4$ for 4+1 points). Thus, function (a) x^8 exhibits error at machine precision even for the largest h considered.

Function (b) is similar to a third-order polynomial, but with an order-3 discontinuity at $x = 1/\sqrt{2}$. Away from the discontinuity, the function behaves as a third-order polynomial and Gaussian quadrature achieves machine precision. Near the discontinuity, the order-3 error shrinks with a smaller cell width h since polynomials increasingly well-approximate the function at high-resolution. Thus, the error decays with shrinking h with a power law relationship approaching h^4.

Similarly, function (c) achieves machine precision away from the discontinuity but has a dominant error determined by the discontinuity. The error from the discontinuity (i.e., quality of polynomial fit) only improves according to decreases in h, implying linear convergence.

Function (d), on the other hand, is smooth but does not resemble a polynomial. Thus, we expect a slow convergence that improves with smaller h but does not inherently capture the behavior of the intended function. This is because $\sqrt{x}$ has a singularity at 0 and does not well approximate to a polynomial (its derivatives contain $1/\sqrt{x}$, so we cannot use Taylor series approximations). We observe convergence following $h^{0.5}$, a relatively slow convergence.

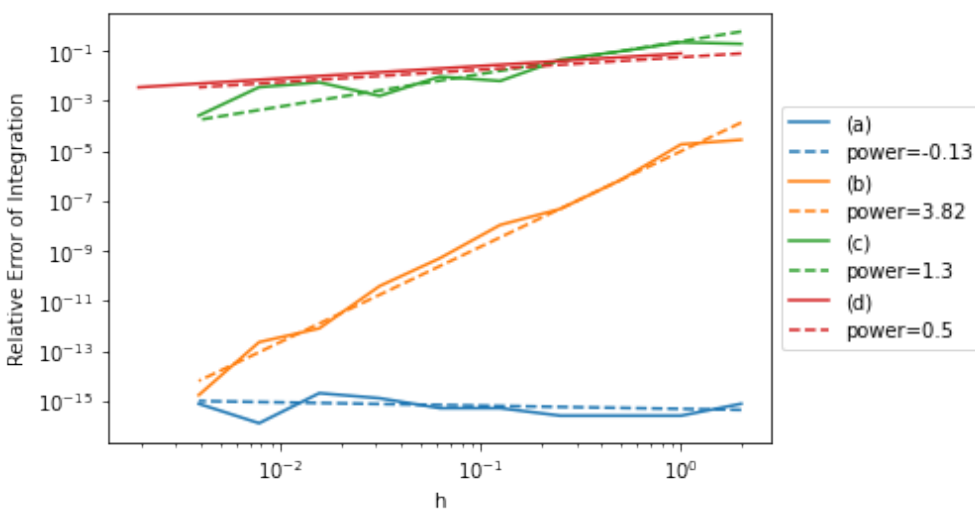


Figure 4: Relative errors of integration for functions (a)-(d) at varied h. Dashed lines indicate power law fit, with legend quantities indicating B in the power law model $A \cdot h^B$.

3 Iterative Methods

Consider the following NxN cyclic tri-diagonal matrix:

$$\mathbb{T}_\alpha = \begin{pmatrix} \alpha & -1 & 0 & ... & -1 \\ -1 & \alpha & -1 & ... & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & ... & \alpha & -1 \\ -1 & 0 & ... & -1 & \alpha \end{pmatrix}$$

(a) *Show that the following vectors (which are discrete Fourier modes)*

$$\mathbf{v}_m = \begin{bmatrix} v_{m,0} \\ v_{m,1} \\ \vdots \\ v_{m,N-1} \end{bmatrix}, \text{ where } v_{m,n} = \exp\left(-i\frac{2\pi mn}{N}\right), n, m = 0, ..., N-1$$

are eigenvectors of $\mathbb{T}_\alpha$ *with eigenvalues given by*

$$\lambda_m = \alpha - 2\cos\left(\frac{2\pi m}{N}\right)$$

The matrix-vector multiplication with the m-eigenvector

$$\mathbb{T}\mathbf{v}_m = \begin{pmatrix} \alpha & -1 & 0 & ... & -1 \\ -1 & \alpha & -1 & ... & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & ... & \alpha & -1 \\ -1 & 0 & ... & -1 & \alpha \end{pmatrix} \begin{bmatrix} 1 \\ \exp\left(-i\frac{2\pi m}{N}\right) \\ \vdots \\ \exp\left(-i\frac{2\pi m(N-1)}{N}\right) \end{bmatrix} = \mathbf{y}$$

results in a vector $\mathbf{y}$ where the ℓth component is

$$y_\ell = \sum_{j=0}^{N-1} \mathbb{T}_{j-\ell} v_j^{(m)} \tag{1}$$

Notice that $j-l$ indexes the position within the matrix with respect to the main diagonal for symmetric matrices.

Equation 1 can be rewritten:

$$\begin{aligned} y_\ell &= \sum_{j=0}^{N-1} \mathbb{T}_{j-\ell} \left[\exp\left(-i\frac{2\pi}{N}\right)\right]^{mj} \\ &= \left[\exp\left(-i\frac{2\pi}{N}\right)\right]^{m\ell} \cdot \sum_{j=0}^{N-1} \mathbb{T}_{j-\ell} \left[\exp\left(-i\frac{2\pi}{N}\right)\right]^{m(j-\ell)} \end{aligned}$$

where the sum is now independent of ℓ. That is, since the summed components are periodic in j, then shifting the index j by l will result in the same total sum.

Therefore,

$$y_\ell = \lambda_m \left[\exp\left(-i\frac{2\pi}{N}\right)\right]^{m\ell} = \lambda_m \mathbf{v}_m, \text{ where } \lambda_m = \sum_{j=0}^{N-1} \mathbb{T}_j \left[\exp\left(-i\frac{2\pi}{N}\right)\right]^{mj} \tag{2}$$

Since the matrix-vector multiplication of $\mathbb{T}$ with $\mathbf{v}_m$ results in a scalar multiple of $\mathbf{v}_m$, we conclude that $\mathbf{v}_m$ is the m-eigenvector with corresponding eigenvalue λ_m.

As the m-eigenvalue is row-independent, we can apply the expression in Equation 2. First, rewrite

$$\omega = \exp\left(-i\frac{2\pi}{N}\right)$$

Then,

$$\begin{aligned}
\lambda_m &= \alpha\omega^0 - \omega^m - \omega^{m(N-1)} \\
&= \alpha - \omega^m - \omega^{mN}\omega^{-m} \\
&= \alpha - \omega^m - (1)^m\omega^{-m} \\
&= \alpha - 2\cos\left(\frac{2\pi m}{N}\right)
\end{aligned}$$

Source: [1]

(b) *Under what condition is the matrix $\mathbb{T}_\alpha$ positive definite?*

The matrix is positive definite if

$$\lambda_m > 0 \text{ and } \lambda_m \in \Re \forall m$$

For this to be true,

$$\alpha > 2\cos\left(\frac{2\pi m}{N}\right)\forall m$$

(c) *Compute the spectral radius of the Jacobi iteration scheme for matrix $\mathbb{T}_\alpha$.*

Decomposing $\mathbb{T}$ into lower, upper, and diagonal parts gives

$$\mathbb{T} = \mathbb{D} + \mathbb{L} + \mathbb{U} \text{ where } \mathbb{D} = \alpha\mathbb{I}$$

$$\therefore \mathbb{D}^{-1} = \frac{1}{\alpha}\mathbb{I}$$

The Jacobi iteration matrix, $\mathbb{B}$, is

$$\begin{aligned}
\mathbb{B} &= \mathbb{D}^{-1}(\mathbb{L} + \mathbb{U}) \\
&= \mathbb{I} - \mathbb{D}^{-1}\mathbb{T} \\
&= \mathbb{I} - \frac{1}{\alpha}\mathbb{T}
\end{aligned}$$

Thus, the eigenvalues of $\mathbb{B}$ are

$$\begin{aligned}
\mu_m &= 1 - \frac{\lambda_m}{\alpha} \\
&= \frac{2}{\alpha}\cos\left(\frac{2\pi m}{N}\right)
\end{aligned}$$

By Taylor series approximation, the spectral radius is therefore

$$\rho = \mu_m\big|_{\max} = \frac{2}{\alpha} - \frac{4\pi^2 k_{\min}^2}{\alpha N^2}$$

indicating convergence for $\rho < 1$. So, the choice of $k_{\min}$ is important.

For $k_{\min} = 0$ (since we are considering $n, m = 0, ..., N-1$),

$$\rho = \frac{2}{\alpha}$$

Source: [2]

(d) *Consider now the specific case $\alpha = 2$. The matrix $\mathbb{T}_\alpha$ now corresponds to the second-order finite difference approximation of Laplace equation. We want to solve the linear system*

$$\mathbb{T}_\alpha \mathbf{x} = \lambda_m \mathbf{v}_m$$

where $\mathbf{v}_m$ is a sine mode. Since a sine mode is a linear combination of two Fourier modes with eigenvalue λ_m, it is also an eigenvector and therefore is a trivial solution. Implement the Jacobi iterative scheme for 1000 iterations to solve for x (using $N = 100$) starting from $\mathbf{x}^{(0)} = 0$.

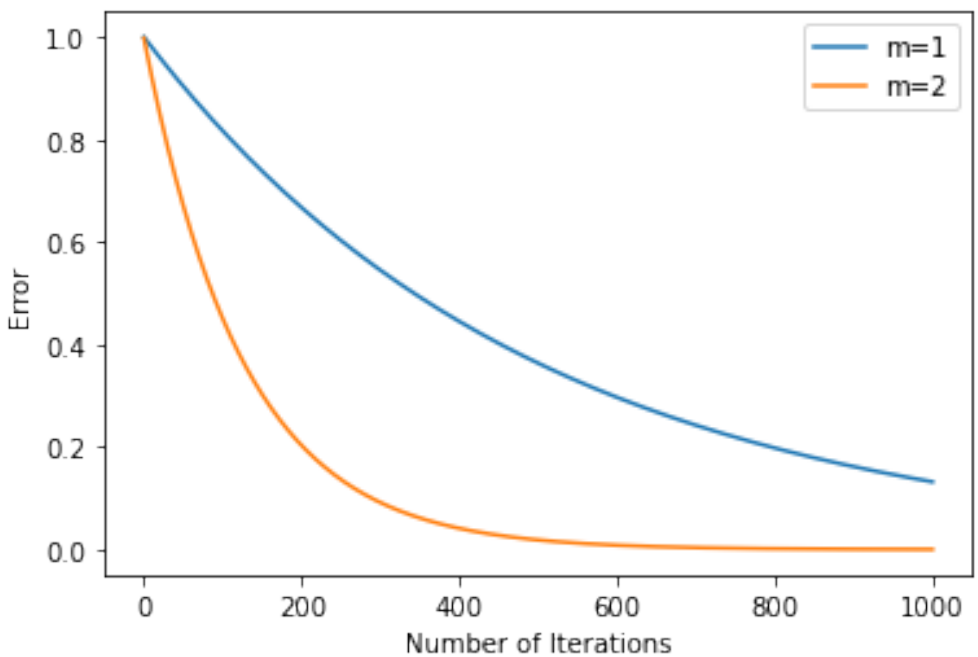


Figure 5: Caption

1. The observed convergence rates are 0.998 and 0.992 for $\mathbf{v}_1$ and $\mathbf{v}_2$, respectively. This was found by fitting an exponential model ($A \cdot B^{iter}$) to each curve in Figure 5.
2. The first mode fits well to the model $(0.998)^{iter}$ and has an exact convergence rate predicted by:

$$\frac{2}{\alpha} \cos\left(\frac{2\pi}{N}\right) \approx 0.998$$

The second mode fits well to the model $(0.992)^{iter}$ and has first-order convergence rate predicted by:

$$\frac{2}{\alpha} \cos\left(\frac{4\pi}{N}\right) \approx 0.992$$

Both observed convergence rates match the predicted values. Notably, the convergence rate of the second mode is faster than the first (the B value in the model B^{iter} is a smaller fraction and thus decays faster), consistent with the predicted faster convergence of the second mode.

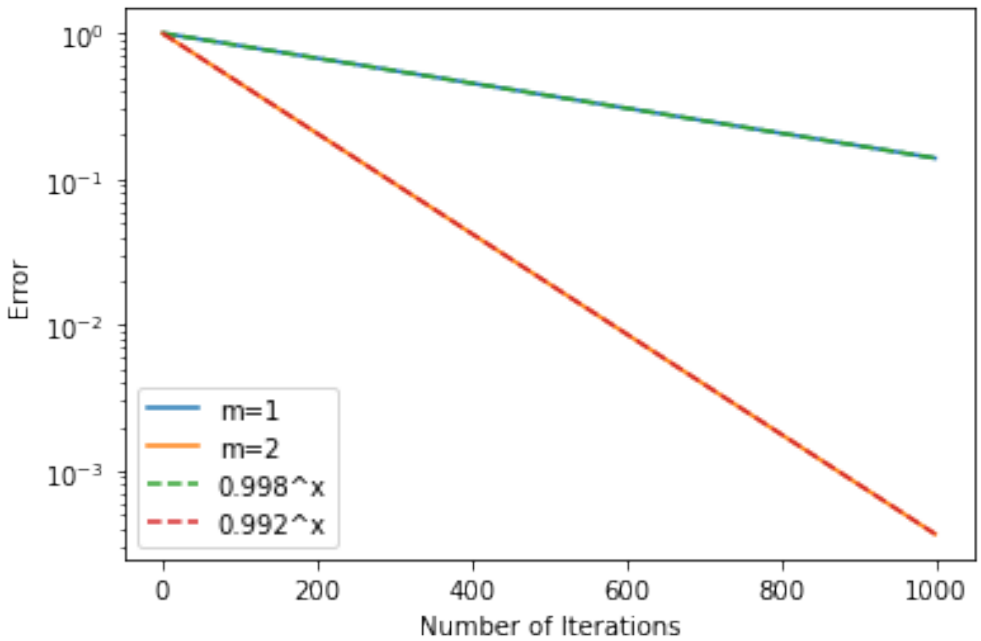


Figure 6: Exponential fits to find convergence rates for modes 1 and 2.

3. For an arbitrary vector, $\mathbf{y}$, that is a linear combination of the set of eigenvectors, I expect the convergence rate to be close to 0.998 (the convergence rate for mode 1). I expect that higher modes will exhibit faster error decay, leaving the error from mode 1 to dominate the convergence behavior.

References

[1] *Circulant Matrices*. 2017. URL: https://web.mit.edu/18.06/www/Spring17/Circulant-Matrices.pdf.

[2] Gilbert Strang. *6.2 Iterative Methods (lecture notes)*. 2006. URL: https://ocw.mit.edu/courses/18-086-mathematical-methods-for-engineers-ii-spring-2006/4bea2ab7b0fa37cd96f2ce442efa769d_am62.pdf.

[3] Lloyd N. Trefethen. *Chebfun and Approximation Theory*. University of Oxford, 2009. URL: https://www.chebfun.org/docs/guide/guide04.html.